

Research suggests that individuals with depression experience dysfunctions in their autobiographical memory. For example, negative autobiographical memories are more easily recalled than positive memories, and memories encoded while experiencing a positive mood are more difficult to access while depressed. In individuals with depression, rumination (repetitive negative thoughts) can also impede autobiographical memory retrieval.

Studies have demonstrated that the Method of Loci (MoL), a mnemonic strategy that pairs items to be remembered with well-known visuospatial locations (loci), effectively improves retention and recall over time. Given this previous research, experimenters hypothesized that training individuals with depression in the MoL technique would improve sustained retrieval of their positive autobiographical memories. For this study, 125 individuals between ages 25 and 45 with a current diagnosis of depression were recruited (individuals with co-occurring psychological diagnoses were excluded). Phase 1 screened participants for normal memory function. Participants were given a list of 25 three-letter nonsense words to memorize (eg, LAR, PAS, FUV), and recall was tested immediately (no delay), and again after 20 minutes, one hour, nine hours, 24 hours, 48 hours, and 72 hours. Those participants who demonstrated normal memory function ($n = 122$) were then randomly assigned to the MoL group ($n = 61$) or the control group ($n = 61$) for the next phase.

During phase 2, the MoL group was trained to associate features of a positive autobiographical memory (eg, a marriage proposal) with a well-known route (eg, the route from home to work). All salient details of the positive memory were mentally associated with loci. For example, one participant paired the restaurant table where he proposed to his wife with a fire hydrant he passes on his way to work. The control group was not trained using any specific technique. Both groups mentally rehearsed their chosen memory every day for one week. After one week, both groups performed equally well at recalling their selected positive memory. However, when a surprise recollection memory test was performed a month later, the MoL group performed at levels comparable to their first recollection test, while the control group exhibited significantly reduced recollection.

Which of the following is the most likely reason the researchers limited the age range of the participants included in this study? Compared to younger adults, older adults experience:

✔ ☒ A. age-related declines in autobiographical memory. [77%]

- ☐ B. enhanced recall of autobiographical events occurring during adolescence and early adulthood. [13%]
- ☐ C. age-related declines in crystallized intelligence. [5%]
- ☐ D. enhanced ability to solve problems and adapt to new situations. [3%]

Correct

77% answered correctly

Explanation:

Age-related cognitive effects in older adults	
Declines	Remains stable
- Memory: episodic (personal autobiographical events) - Intelligence: fluid (ability to creatively solve new problems & see new patterns)	- Memory: semantic (fact-based) & procedural (skills-based) - Intelligence: crystallized (ability to apply established skills or knowledge)

Older adults experience age-related declines in certain cognitive functions. Research has shown there are age-related **declines** in **episodic memory**, but **semantic** (fact-based) and **procedural** (skill-based) memories appear to remain relatively **stable**. Research has also demonstrated age-related **declines** in **fluid** (adaptable) **intelligence**, but **crystallized** (acquired skills/knowledge) **intelligence** appears to remain relatively **unchanged** and even increases with age (to a point).

Because the ability to recall an episodic (autobiographical) memory was the dependent variable in this study, the researchers most likely limited the age range of the participants to 25-45 years because age-related declines in autobiographical (ie, episodic) memory in older adults could impact the study results.

(Choice B) Older adults tend to recall events from their youth and younger adulthood more easily than events that happened later; this is termed the reminiscence bump. The reminiscence bump describes older adults' enhanced recall *compared to their own recall* of other memories, not compared to younger adults.

(Choice C) Crystallized intelligence, the ability to apply established skills or knowledge, is thought to remain stable (or increase) with age, not decline.

(Choice D) Fluid intelligence, the ability to use logic and creativity to solve new problems and identify new patterns, peaks in early adulthood and declines with age.

Educational objective:

Aging has been associated with declines in episodic memory and fluid intelligence. Semantic and procedural memory appear to remain relatively stable with age; crystallized intelligence appears to remain stable (or increase).

Time spent:0 sec**Subject:**

Behavioral Sciences

QID:401997**System:**

Learning, Memory, and Cognition

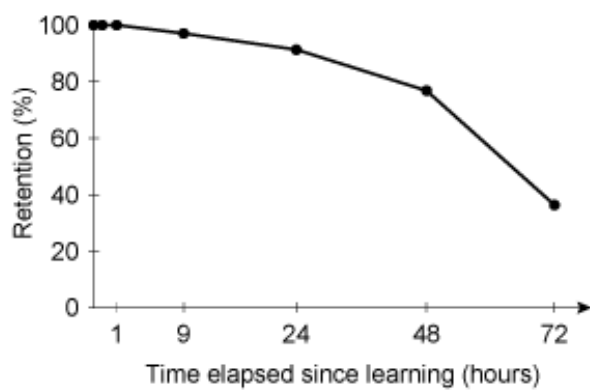
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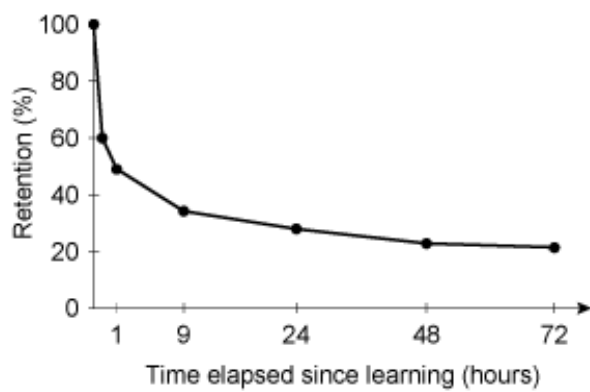
Which of the following hypothetical Phase 1 results best illustrates normal memory decay?

☐ A.



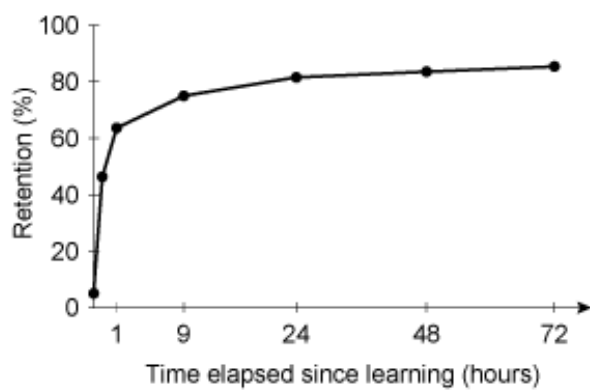
[40%]

✓ ☒ B.

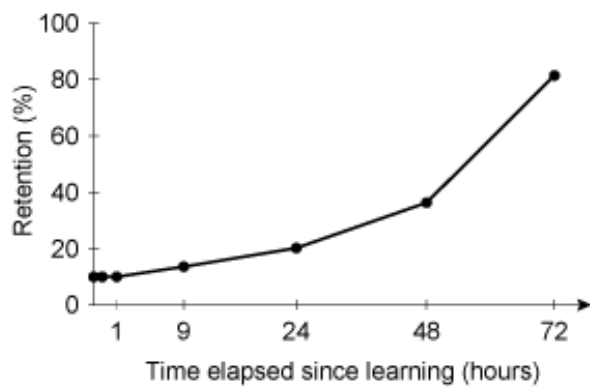


[57%]

☐ C.



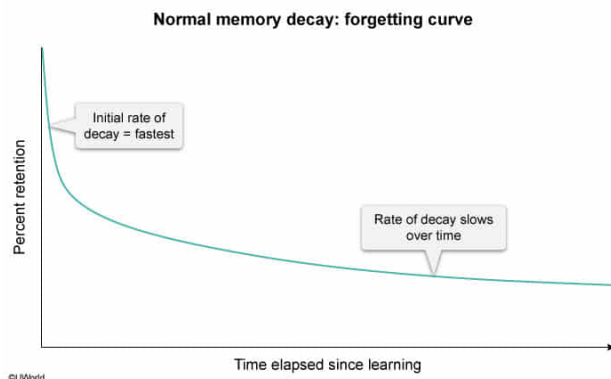
☐ D.



Correct

58% answered correctly

Explanation:



Phase 1 of the study was designed to screen participants to ensure that those included in the study demonstrated normal memory function. For this test, participants were asked to remember 25 three-letter nonsense words, with their retention tested immediately and after 20 minutes, one hour, nine hours, 24 hours, 48 hours, and 72 hours.

Normal **memory decay** (or **forgetting**) follows a characteristic pattern: the **initial rate of decay is highest** right after the material is first learned, then plateaus over time unless the material is reviewed. Numerous studies have produced the same basic forgetting curve shape for different types of memory, including short-term and several types of long-term memory (eg, episodic, semantic, and procedural).

These graphs depict percent retention (amount remembered) on the y-axis, as a function of the time elapsed since learning, on the x-axis. Normal memory decay should be depicted by a steep initial drop in percent retained that plateaus over time, as does the graph in [Choice B](#).

(Choice A) Retention remaining very high and stable over the first few hours after learning is *not typical* and does not demonstrate normal memory decay.

(Choices C and D) In the absence of techniques to improve retention (eg, review of the material), normal memory retention does not *increase* over time. Normal memory decay tends to decrease in a characteristic pattern where a large proportion of the material is initially lost, and then the rate of decay slows over time.

Educational objective:

Normal memory decay (or forgetting) follows a characteristic pattern: the initial rate of decay is highest right after the material is first learned, then plateaus over time.

Time spent:0 sec
Subject:
Behavioral Sciences

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System:
Learning, Memory, and Cognition